

Question ID: beca03de

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

A rectangle has a length that is **15** times its width. The function $y = (15w)(w)$ represents this situation, where y is the area, in square feet, of the rectangle and $y > 0$. Which of the following is the best interpretation of **15w** in this context?

Answer

- A. The length of the rectangle, in feet
- B. The area of the rectangle, in square feet
- C. The difference between the length and the width of the rectangle, in feet
- D. The width of the rectangle, in feet

Question ID: 4e18fc5d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$$v = -\frac{w}{150x}$$

The given equation relates the distinct positive numbers v , w , and x . Which equation correctly expresses w in terms of v and x ?

Answer

- A. $w = -150vx$
- B. $w = -\frac{150v}{x}$
- C. $w = -\frac{x}{150v}$
- D. $w = v + 150x$

Question ID: f5c3e3b8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question

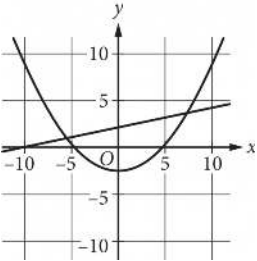
Which expression is equivalent to $(m^4q^4z^{-1})(mq^5z^3)$, where m , q , and z are positive?

Answer

- A. $m^4q^{20}z^{-3}$
- B. $m^5q^9z^2$
- C. $m^6q^8z^{-1}$
- D. $m^{20}q^{12}z^{-2}$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question



A system of equations consists of a quadratic equation and a linear equation. The equations in this system are graphed in the xy -plane above. How many solutions does this system have?

Answer

- A. 0
- B. 1
- C. 2
- D. 3

Question ID: 3c95093c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question
 $6x - 9y > 12$

Which of the following inequalities is equivalent to the inequality above?

- Answer
- A. $x - y > 2$
- B. $2x - 3y > 4$
- C. $3x - 2y > 4$
- D. $3y - 2x > 2$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$y = x + 1$
 $y = x^2 + x$

If (x,y) is a solution to the system of equations above, which of the following could be the value of x ?

Answer

- A. -1
- B. 0
- C. 2
- D. 3

Question ID: 72ebc024

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question

Which expression is equivalent to $16x^3y^2 + 14xy$?

Answer

- A. $2xy(8xy + 7)$
- B. $2xy(8x^2y + 7)$
- C. $14xy(2x^2y + 1)$
- D. $14xy(8x^2y + 1)$

Question ID: dd4ab4c4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

$4a^2 + 20ab + 25b^2$

Which of the following is a factor of the polynomial above?

Answer

- A. $a + b$
- B. $2a + 5b$
- C. $4a + 5b$
- D. $4a + 25b$

Question ID: b8caaf84

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

If $p = 3x + 4$ and $v = x + 5$, which of the following is equivalent to $pv - 2p + v$?

Answer

- A. $3x^2 + 12x + 7$
- B. $3x^2 + 14x + 17$
- C. $3x^2 + 19x + 20$
- D. $3x^2 + 26x + 33$

Question ID: f01108a8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$y = 9\left(\frac{a}{7}\right)^{x+c} - b$$

How many times does the graph of the given equation in the xy -plane cross the x -axis, where a, b , and c are positive constants such that $a > 7$ and $b > c$?

Answer

- A. Zero
- B. One
- C. Two
- D. Three

Question ID: 7f81d0c3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$x^2 - x - 1 = 0$

What values satisfy the equation above?

Answer

A. $x = 1$ and $x = 2$

B. $x = -\frac{1}{2}$ and $x = \frac{3}{2}$

C. $x = \frac{1 + \sqrt{5}}{2}$ and $x = \frac{1 - \sqrt{5}}{2}$

D. $x = \frac{-1 + \sqrt{5}}{2}$ and $x = \frac{-1 - \sqrt{5}}{2}$

Question ID: 332cd67b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question

$3x^2 - 15x + 18 = 0$

How many distinct real solutions are there to the given equation?

Answer

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

Question ID: 301faf80

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The product of two positive integers is **462**. If the first integer is **5** greater than twice the second integer, what is the smaller of the two integers?

Question ID: 128c75e2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function g is defined by $g(x) = \frac{|x|}{a} - 14$, where $a < 0$. What is the product of $g(15a)$ and $g(7a)$?

Question ID: 71b74a44

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

$y = 7,400(0.87)^x$

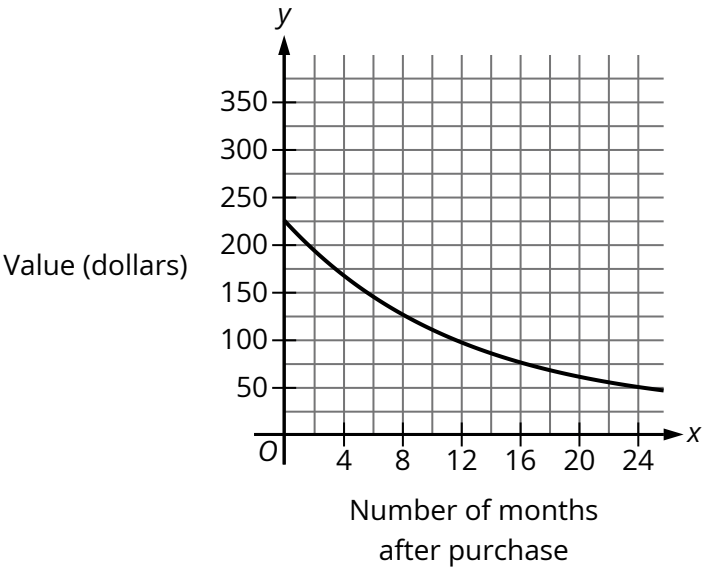
The given equation estimates the value, in dollars, of a certain piece of equipment x years after it was purchased. According to the given equation, what was the estimated value, in dollars, of the piece of equipment at the time it was purchased?

Answer

- A. 740
- B. 6,438
- C. 7,400
- D. 8,700

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question



The graph shown gives the estimated value, in dollars, of a tablet as a function of the number of months since it was purchased. What is the best interpretation of the y -intercept of the graph in this context?

Answer

- A. The estimated value of the tablet was **\$225** when it was purchased.
- B. The estimated value of the tablet **24** months after it was purchased was **\$225**.
- C. The estimated value of the tablet had decreased by **\$225** in the **24** months after it was purchased.
- D. The estimated value of the tablet decreased by approximately **2.25%** each year after it was purchased.

Question ID: e312081b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question
 $(x + 5) + (2x - 3)$

Which of the following is equivalent to the given expression?

Answer

- A. $3x - 2$
- B. $3x + 2$
- C. $3x - 8$
- D. $3x + 8$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

x	$g(x)$
-27	3
-9	0
21	5

The table shows three values of x and their corresponding values of $g(x)$, where $g(x) = \frac{f(x)}{x+3}$ and f is a linear function. What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

- Answer
- A. $(0, 36)$
 - B. $(0, 12)$
 - C. $(0, 4)$
 - D. $(0, -9)$

Question ID: 52931bfa

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

Which expression is equivalent to $\frac{8x(x-7)-3(x-7)}{2x-14}$, where $x > 7$?

Answer

- A. $\frac{x-7}{5}$
- B. $\frac{8x-3}{2}$
- C. $\frac{8x^2-3x-14}{2x-14}$
- D. $\frac{8x^2-3x-77}{2x-14}$

Question ID: 1e003284

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question

$x = 49$

$y = \sqrt{x} + 9$

The graphs of the given equations intersect at the point (x, y) in the xy -plane. What is the value of y ?

Answer

- A. 16
- B. 40
- C. 81
- D. 130

Question ID: 91e7ea5e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$h(x) = 2(x - 4)^2 - 32$

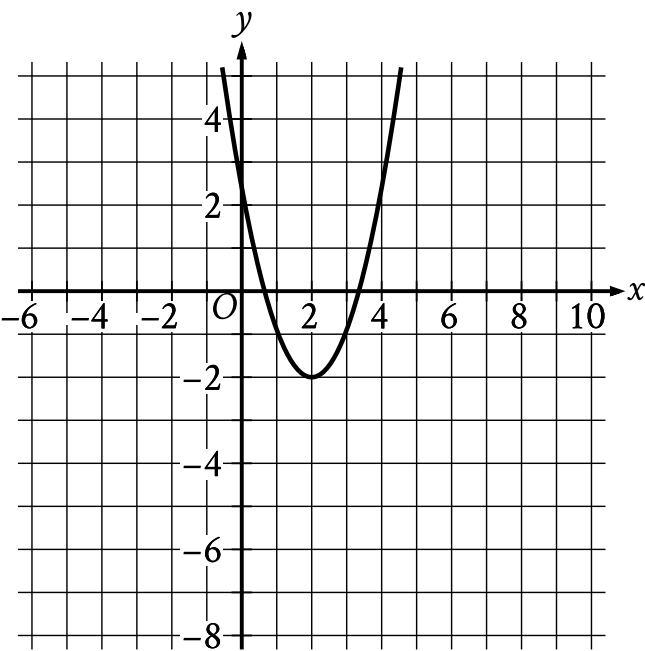
The quadratic function h is defined as shown. In the xy-plane, the graph of $y = h(x)$ intersects the x-axis at the points $(0,0)$ and $(t,0)$, where t is a constant. What is the value of t ?

Answer

- A. 1
- B. 2
- C. 4
- D. 8

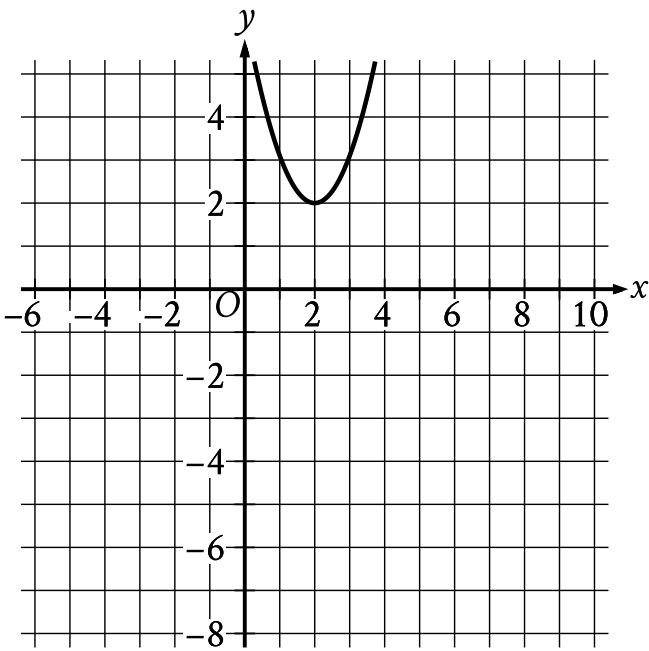
Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

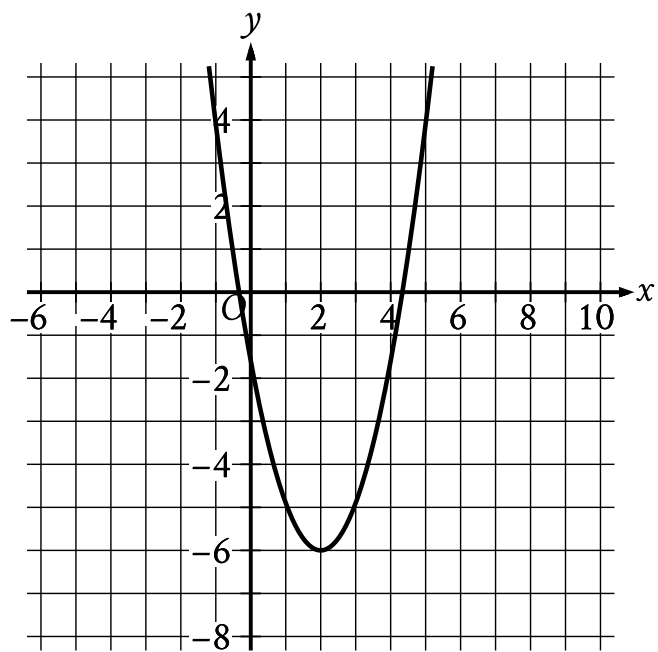


The graph shown will be translated up 4 units. Which of the following will be the resulting graph?

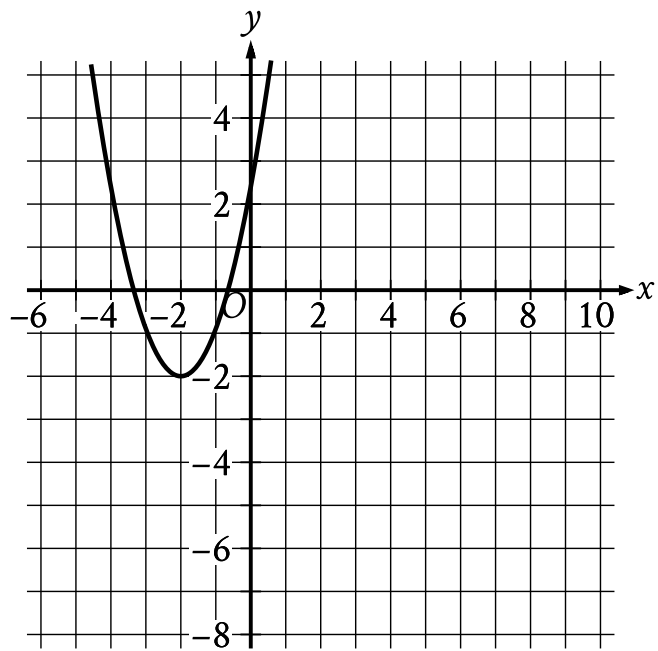
Answer



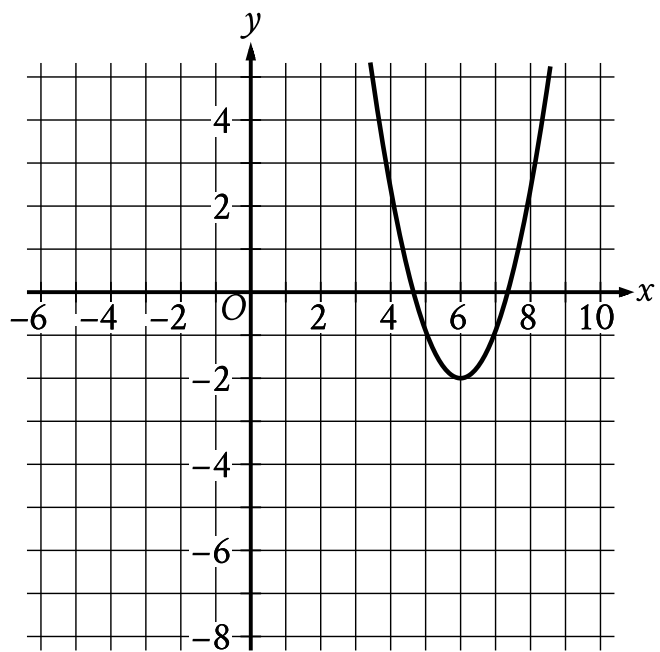
A.



B.



C.



D.

Question ID: 358f18bc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$f(x) = x^2 - 48x + 2,304$

What is the minimum value of the given function?

Question ID: 3a9d60b2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$2|4 - x| + 3|4 - x| = 25$

What is the positive solution to the given equation?

Question ID: 8490cc45

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

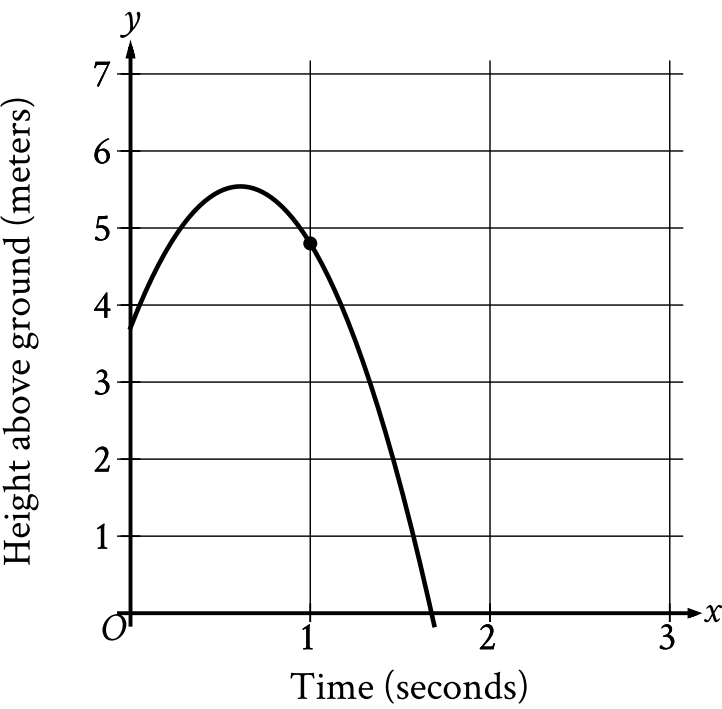
The function f is defined by $f(x) = (-8)(2)^x + 22$. What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

Answer

- A. $(0, 14)$
- B. $(0, 2)$
- C. $(0, 22)$
- D. $(0, -8)$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question



The graph shows the height above ground, in meters, of a ball x seconds after the ball was launched upward from a platform. Which statement is the best interpretation of the marked point $(1.0, 4.8)$ in this context?

Answer

- A. **1.0** second after being launched, the ball's height above ground is **4.8** meters.
- B. **4.8** seconds after being launched, the ball's height above ground is **1.0** meter.
- C. The ball was launched from an initial height of **1.0** meter with an initial velocity of **4.8** meters per second.
- D. The ball was launched from an initial height of **4.8** meters with an initial velocity of **1.0** meter per second.

Choice D is incorrect and may result from conceptual errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

An auditorium has seats for **1,800** people. Tickets to attend a show at the auditorium currently cost **\$4.00**. For each **\$1.00** increase to the ticket price, **100** fewer tickets will be sold. This situation can be modeled by the equation $y = -100x^2 + 1,400x + 7,200$, where x represents the increase in ticket price, in dollars, and y represents the revenue, in dollars, from ticket sales. If this equation is graphed in the xy -plane, at what value of x is the maximum of the graph?

Answer

- A. **4**
- B. **7**
- C. **14**
- D. **18**

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$x^2 - 2x - 9 = 0$

One solution to the given equation can be written as $1 + \sqrt{k}$, where k is a constant. What is the value of k ?

Answer

- A. 8
- B. 10
- C. 20
- D. 40

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

$p(x) + 57 = x^2$

The given equation relates the value of x and its corresponding value of $p(x)$ for the function p . What is the minimum value of the function p ?

Answer

- A. ~~−3,249~~
- B. ~~−57~~
- C. ~~57~~
- D. ~~3,249~~

Question ID: fc3d783a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

In the xy -plane, a line with equation $2y = 4.5$ intersects a parabola at exactly one point. If the parabola has equation $y = -4x^2 + bx$, where b is a positive constant, what is the value of b ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$f(x) = 9,000(0.66)^x$

The given function f models the number of advertisements a company sent to its clients each year, where x represents the number of years since **1997**, and $0 \leq x \leq 5$. If $y = f(x)$ is graphed in the xy -plane, which of the following is the best interpretation of the y -intercept of the graph in this context?

Answer

- A. The minimum estimated number of advertisements the company sent to its clients during the **5** years was **1,708**.
- B. The minimum estimated number of advertisements the company sent to its clients during the **5** years was **9,000**.
- C. The estimated number of advertisements the company sent to its clients in **1997** was **1,708**.
- D. The estimated number of advertisements the company sent to its clients in **1997** was **9,000**.

Question ID: 075b29b0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question

Which expression is equivalent to $(9x^3 + 5x + 7) + (6x^3 + 5x^2 - 5)$?

Answer

- A. $15x^6 + 5x^2 - 5x - 35$
- B. $15x^3 + 10x^2 + 2$
- C. $15x^6 + 5x^2 + 5x + 2$
- D. $15x^3 + 5x^2 + 5x + 2$

Question ID: 2c6f214f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The first term of a sequence is **9**. Each term after the first is **4** times the preceding term. If *w* represents the *n*th term of the sequence, which equation gives *w* in terms of *n*?

Answer

- A. $w = 4(9^n)$
- B. $w = 4(9^{n-1})$
- C. $w = 9(4^n)$
- D. $w = 9(4^{n-1})$

Question ID: 781c2f6e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function f is defined by $f(x) = a(2.2^x + 2.2^b)$, where a and b are integer constants and $0 < a < b$. The functions g and h are equivalent to function f , where k and m are constants. Which of the following equations displays the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane as a constant or coefficient?

- I. $g(x) = a(2.2^x + k)$
- II. $h(x) = a(2.2)^x + m$

Answer

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Question ID: cb29c54c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

A physics class is planning an experiment about a toy rocket. The equation $y = -16(x - 5.6)^2 + 502$ gives the estimated height y , in feet, of the toy rocket x seconds after it is launched into the air. Which of the following is the best interpretation of the vertex of the graph of the equation in the xy -plane?

Answer

- A. This toy rocket reaches an estimated maximum height of **502** feet **16** seconds after it is launched into the air.
- B. This toy rocket reaches an estimated maximum height of **502** feet **5.6** seconds after it is launched into the air.
- C. This toy rocket reaches an estimated maximum height of **16** feet **502** seconds after it is launched into the air.
- D. This toy rocket reaches an estimated maximum height of **5.6** feet **502** seconds after it is launched into the air.

Question ID: 4661e2a9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$x - y = 1$
 $x + y = x^2 - 3$

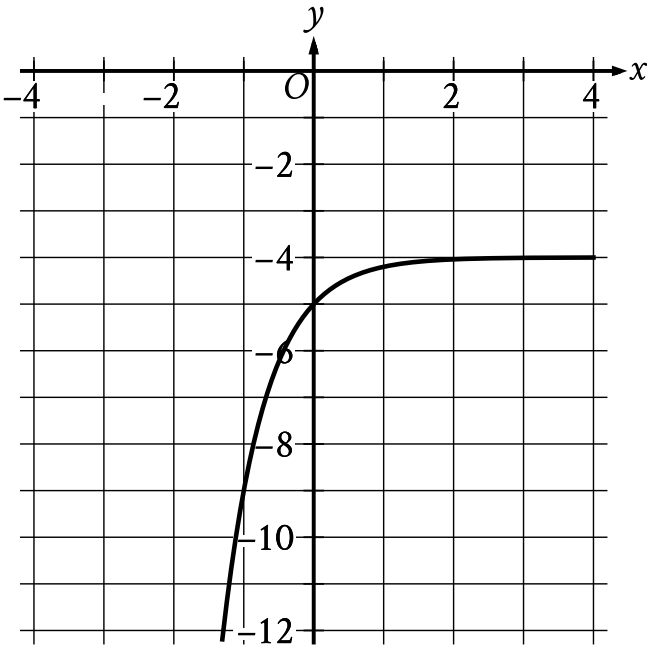
Which ordered pair is a solution to the system of equations above?

Answer

- A. $(1 + \sqrt{3}, \sqrt{3})$
- B. $(\sqrt{3}, -\sqrt{3})$
- C. $(1 + \sqrt{5}, \sqrt{5})$
- D. $(\sqrt{5}, -1 + \sqrt{5})$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question



What is the y -intercept of the graph shown?

- Answer
- A. $(-1, -9)$
 - B. $(0, -5)$
 - C. $(0, -4)$
 - D. $(0, 0)$

Question ID: ad2ec615

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

Which of the following is equivalent to the expression $x^4 - x^2 - 6$?

Answer

- A. $(x^2 + 1)(x^2 - 6)$
- B. $(x^2 + 2)(x^2 - 3)$
- C. $(x^2 + 3)(x^2 - 2)$
- D. $(x^2 + 6)(x^2 - 1)$

Question ID: 42c71eb5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

$(2x + 5)^2 - (x - 2) + 2(x + 3)$

Which of the following is equivalent to the expression above?

Answer

- A. $4x^2 + 21x + 33$
- B. $4x^2 + 21x + 29$
- C. $4x^2 + x + 29$
- D. $4x^2 + x + 33$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

Time (years)	Total amount (dollars)
0	604.00
1	606.42
2	608.84

Rosa opened a savings account at a bank. The table shows the exponential relationship between the time t , in years, since Rosa opened the account and the total amount n , in dollars, in the account. If Rosa made no additional deposits or withdrawals, which of the following equations best represents the relationship between t and n ?

Answer

- A. $n = (1 + 604)^t$
- B. $n = (1 + 0.004)^t$
- C. $n = 604(1 + 0.004)^t$
- D. $n = 0.004(1 + 604)^t$

Question ID: 371cbf6b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$(ax + 3)(5x^2 - bx + 4) = 20x^3 - 9x^2 - 2x + 12$

The equation above is true for all x, where a and b are constants. What is the value of ab ?

Answer

- A. 18
- B. 20
- C. 24
- D. 40

Question ID: b4acba95

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$$x^2 - 12x + 27 = 0$$

How many distinct real solutions does the given equation have?

Answer

- A. Exactly two
- B. Exactly one
- C. Zero
- D. Infinitely many

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

x	$g(x)$
-1	25
0	1
1	$\frac{1}{25}$
2	$\frac{1}{625}$

For the exponential function g , the table shows four values of x and their corresponding values of $g(x)$. Which equation defines g ?

Answer

- A. $g(x) = -25^x$
- B. $g(x) = -\left(\frac{1}{25}\right)^x$
- C. $g(x) = 25^x$
- D. $g(x) = \left(\frac{1}{25}\right)^x$

Question ID: a05bd3a4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

Which of the following expressions is equivalent to $x^2 - 5$?

Answer

A. $(x + \sqrt{5})^2$

B. $(x - \sqrt{5})^2$

C. $(x + \sqrt{5})(x - \sqrt{5})$

D. $(x + 5)(x - 1)$

Question ID: c3b116d7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

Which of the following expressions is(are) a factor of $3x^2 + 20x - 63$?

- I. $x - 9$
- II. $3x - 7$

Answer

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Question ID: 40c09d66

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

If $\frac{\sqrt{x^5}}{\sqrt[3]{x^4}} = x^{\frac{a}{b}}$ for all positive values of x, what is the value of $\frac{a}{b}$?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

x	$h(x)$
0	1.23
2	1.54
4	1.94

The table shows the exponential relationship between the number of years, x , since Hana started training in pole vault, and the estimated height $h(x)$, in meters, of her best pole vault for that year. Which of the following functions best represents this relationship, where $x \leq 4$?

Answer

- A. $h(x) = 1.12(0.23)^x$
- B. $h(x) = 1.12(1.23)^x$
- C. $h(x) = 1.23(0.12)^x$
- D. $h(x) = 1.23(1.12)^x$

Question ID: b8f13a3a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

Function f is defined by $f(x) = -a^x + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x) - 12$ has a y -intercept at $(0, -\frac{75}{7})$. The product of a and b is $\frac{320}{7}$. What is the value of a ?

Question ID: 8e1da169

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$f(x) = (x - 44)(x - 46)$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

Answer

- A. 46
- B. 45
- C. 44
- D. −1

Question ID: 1d3c5c95

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

$f(x) = 4,000(0.75)^x$

An entomologist recommended a program to reduce a certain invasive beetle population in an area. The given function estimates this beetle species' population x years after **2012**, where $x \leq 7$. Which of the following is the best interpretation of **4,000** in this context?

Answer

- A. The estimated initial beetle population for this species and area in **2012**
- B. The estimated beetle population for this species and area **7** years after **2012**
- C. The estimated percent decrease in the beetle population for this species and area each year after **2012**
- D. The estimated percent decrease in the beetle population for this species and area every **7** years after **2012**

Question ID: 46308566

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$r^2 + qr = 2r - 55$$

In the given equation, q is an integer constant. The given equation has no real solutions. What is the largest possible value of q ?

Question ID: ae05d37b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

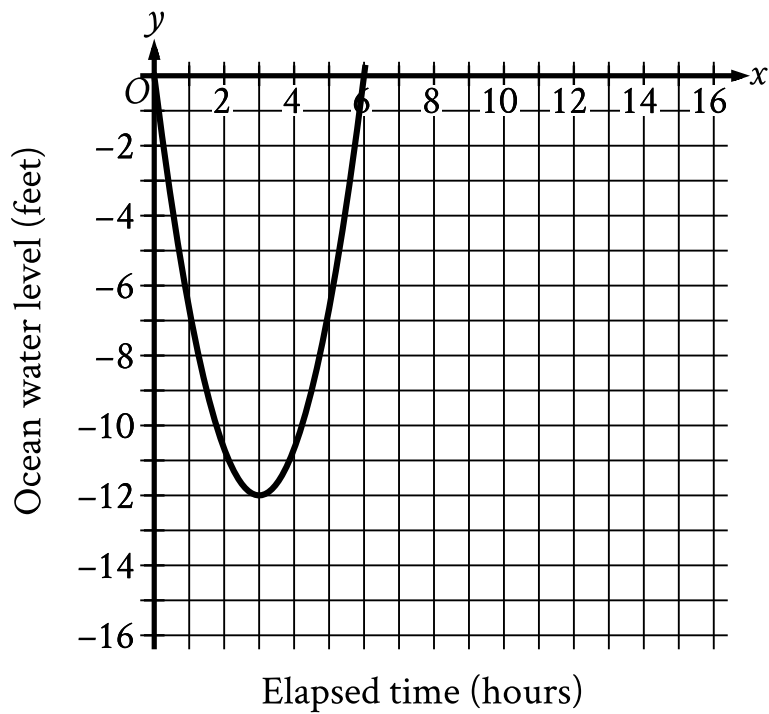
The function $f(t) = 40,000(2)^{\frac{t}{790}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

Answer

- A. 2
- B. 790
- C. 1,580
- D. 40,000

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question



Scientists recorded data about the ocean water levels at a certain location over a period of **6** hours. The graph shown models the data, where $y = 0$ represents sea level. Which table gives values of x and their corresponding values of y based on the model?

Answer

A.

x	y
0	-12
0	3
3	6

B.

x	y
0	0
3	12
0	-6

C.

x	y
0	0
3	-12
6	0

D.

x	y
0	0
12	3

-6	0
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Question ID: c6a26e14

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question

$|x + 45| = 48$

What is the positive solution to the given equation?

Answer

- A. 3
- B. 48
- C. 93
- D. 96

Question ID: 07bcecac

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

$$P(t) = 24.8(1.036)^t$$

The function P gives the predicted population, in millions, of a certain country for the period from **1984** to **2018**, where t is the number of years after **1984**. According to the model, what is the best interpretation of the statement “ $P(8)$ is approximately equal to **32.91**”?

Answer

- A. In **1984**, the predicted population of this country was approximately **8** million.
- B. In **1984**, the predicted population of this country was approximately **32.91** million.
- C. **8** years after **1984**, the predicted population of this country was approximately **32.91** million.
- D. **32.91** years after **1984**, the predicted population of this country was approximately **8** million.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question

$6r = 7s + t$

The given equation relates the variables r , s , and t . Which equation correctly expresses s in terms of r and t ?

Answer

- A. $s = 42r - t$
- B. $s = 7(6r - t)$
- C. $s = \frac{6}{7}r - t$
- D. $s = \frac{6r - t}{7}$

Question ID: 02add2d2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

A company has a newsletter. In January **2018**, there were **1,300** customers subscribed to the newsletter. For the next **24** months after January **2018**, the total number of customers subscribed to the newsletter each month was **7%** greater than the total number subscribed the previous month. Which equation gives the total number of customers, ***c***, subscribed to the company's newsletter ***m*** months after January **2018**, where ***m* ≤ 24**?

Answer

- A. $c = 1,300(0.07)^m$
- B. $c = 1,300(1.07)^m$
- C. $c = 1,300(1.7)^m$
- D. $c = 1,300(7)^m$

Question ID: f65288e8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$\frac{1}{x^2+10x+25} = 4$

If x is a solution to the given equation, which of the following is a possible value of $x+5$?

Answer

- A. $\frac{1}{2}$
- B. $\frac{5}{2}$
- C. $\frac{9}{2}$
- D. $\frac{11}{2}$

Question ID: 788bfd56

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

The function f is defined by $f(x) = 4 + \sqrt{x}$. What is the value of $f(144)$?

Answer

- A. 0
- B. 16
- C. 40
- D. 76

Question ID: 2c288148

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$\sqrt{k - x} = 58 - x$$

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?

Question ID: 40491607

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$f(x) = (x - 1)(x + 3)(x - 2)$

In the xy -plane, when the graph of the function f , where $y = f(x)$, is shifted up **6** units, the resulting graph is defined by the function g . If the graph of $y = g(x)$ crosses through the point $(4, b)$, where b is a constant, what is the value of b ?

Question ID: 369b7bb7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

The function g is defined by $g(x) = \sqrt{8x + 1}$. What is the value of $g(3)$?

Answer

- A. $\frac{5}{8}$
- B. $\frac{25}{8}$
- C. 5
- D. 25

Question ID: 3206b905

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

Which of the following expressions is equivalent to $8x^{10} - 8x^9 + 88x$?

Answer

- A. $x(7x^{10} - 7x^9 + 87x)$
- B. $x(8^{10} - 8^9 + 88)$
- C. $8x(x^{10} - x^9 + 11x)$
- D. $8x(x^9 - x^8 + 11)$

Question ID: f89af023

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question
A rectangular volleyball court has an area of 162 square meters. If the length of the court is twice the width, what is the width of the court, in meters?

- Answer**
- A. 9
 - B. 18
 - C. 27
 - D. 54

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

A machine launches a softball from ground level. The softball reaches a maximum height of **51.84** meters above the ground at **1.8** seconds and hits the ground at **3.6** seconds. Which equation represents the height above ground ***h***, in meters, of the softball ***t*** seconds after it is launched?

- Answer**
- A. $h = -t^2 + 3.6$
 - B. $h = -t^2 + 51.84$
 - C. $h = -16(t - 1.8)^2 - 3.6$
 - D. $h = -16(t - 1.8)^2 + 51.84$

Question ID: 38c90632

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

The expression $\frac{29x+102}{x(x+51)}$ is equivalent to $\frac{p}{x} + \frac{w}{x+51}$, where p and w are constants. What is the value of w ?

Question ID: 4a0d0399

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function f is defined by $f(x) = a^x + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x)$ has an x -intercept at $(2, 0)$ and a y -intercept at $(0, -323)$. What is the value of b ?

Question ID: 3cf2698e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

The number of cells of a certain bacteria doubles every **25** minutes during its growth period when placed in a medium of milk in a controlled laboratory environment. If a sample of **830** cells of this bacteria is placed in milk in a controlled laboratory environment, which function ***f*** best represents the number of cells of the bacteria ***x*** minutes after the growth period begins, where ***x*** is less than the number of minutes in the growth period?

Answer

- A. $f(x) = 830\left(\frac{1}{2}\right)^{\frac{x}{25}}$
- B. $f(x) = 830(2)^{\frac{x}{25}}$
- C. $f(x) = 830\left(\frac{1}{3}\right)^{\frac{x}{25}}$
- D. $f(x) = 830(3)^{\frac{x}{25}}$

Question ID: 768b60d2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

For the exponential function f , the value of $f(0)$ is c , where c is a constant. Of the following equations that define the function f , which equation shows the value of c as the coefficient or the base?

Answer

- A. $f(x) = 22(1.5)^{x+1}$
- B. $f(x) = 33(1.5)^x$
- C. $f(x) = 49.5(1.5)^{x-1}$
- D. $f(x) = 74.25(1.5)^{x-2}$

Question ID: e53add44

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question
 $S(n) = 38,000a^n$

The function S above models the annual salary, in dollars, of an employee n years after starting a job, where a is a constant. If the employee's salary increases by 4% each year, what is the value of a ?

- Answer
- A. 0.04
 - B. 0.4
 - C. 1.04
 - D. 1.4

Question ID: f2f3fa00

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

During a 5-second time interval, the average acceleration a , in meters per second squared, of an object with an initial velocity of 12 meters per second is defined by the equation $a = \frac{v_f - 12}{5}$, where v_f is the final velocity of the object in meters per second. If the equation is rewritten in the form $v_f = xa + y$, where x and y are constants, what is the value of x ?

Question ID: b4a6ed81

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

The expression $90y^5 - 54y^4$ is equivalent to $ry^4(15y - 9)$, where r is a constant. What is the value of r ?

Question ID: 9654add7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$f(x) = -500x^2 + 25,000x$

The revenue $f(x)$, in dollars, that a company receives from sales of a product is given by the function f above, where x is the unit price, in dollars, of the product. The graph of $y = f(x)$ in the xy -plane intersects the x -axis at 0 and a . What does a represent?

Answer

- A. The revenue, in dollars, when the unit price of the product is \$0
- B. The unit price, in dollars, of the product that will result in maximum revenue
- C. The unit price, in dollars, of the product that will result in a revenue of \$0
- D. The maximum revenue, in dollars, that the company can make

Question ID: 4618501a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

$f(x) = 3,000(0.75)^x$

A conservation scientist implemented a program to reduce the population of a certain species in an area. The given function estimates this species' population x years after **2008**, where $x \leq 8$. Which of the following is the best interpretation of **3,000** in this context?

Answer

- A. The estimated percent decrease in the population for this species and area every **8** years after **2008**
- B. The estimated percent decrease in the population for this species and area each year after **2008**
- C. The estimated population for this species and area **8** years after **2008**
- D. The estimated initial population for this species and area in **2008**

Question ID: 34847f8a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$$\frac{2}{x-2} + \frac{3}{x+5} = \frac{rx+t}{(x-2)(x+5)}$$

The equation above is true for all $x > 2$, where r and t are positive constants. What is the value of rt ?

Answer

- A. -20
- B. 15
- C. 20
- D. 60

Question ID: 253985c2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question

Which expression is equivalent to $6x + 5x + 4y$?

Answer

- A. $15x$
- B. $15y$
- C. $11x + 4y$
- D. $30x + 4y$

Question ID: cc776a04

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

Which of the following is an equivalent form of $(1.5x - 2.4)^2 - (5.2x^2 - 6.4)$?

Answer

- A. $-2.2x^2 + 1.6$
- B. $-2.2x^2 + 11.2$
- C. $-2.95x^2 - 7.2x + 12.16$
- D. $-2.95x^2 - 7.2x + 0.64$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

Growth of a Culture of Bacteria

Day	Number of bacteria per milliliter at end of day
1	2.5×10^5
2	5.0×10^5
3	1.0×10^6

A culture of bacteria is growing at an exponential rate, as shown in the table above. At this rate, on which day would the number of bacteria per milliliter reach 5.12×10^8 ?

- Answer
- A. Day 5
 - B. Day 9
 - C. Day 11
 - D. Day 12

Question ID: 4ac59df6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question

Which expression is equivalent to $(8yz)(y)(7z)$?

Answer

- A. $56y^2z^2$
- B. $56y^2z$
- C. $56yz$
- D. $16yz$

Question ID: fada6b03

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$2x^2 - 8x - 7 = 0$

One solution to the given equation can be written as $\frac{8-\sqrt{k}}{4}$, where k is a constant. What is the value of k ?

Question ID: 926c246b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

$D = 5,640(1.9)^t$

The equation above estimates the global data traffic D, in terabytes, for the year that is t years after 2010. What is the best interpretation of the number 5,640 in this context?

Answer

- A. The estimated amount of increase of data traffic, in terabytes, each year
- B. The estimated percent increase in the data traffic, in terabytes, each year
- C. The estimated data traffic, in terabytes, for the year that is t years after 2010
- D. The estimated data traffic, in terabytes, in 2010

Question ID: 84e5e36c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question

$y = 76$

$y = x^2 - 5$

The graphs of the given equations in the xy -plane intersect at the point (x, y) . What is a possible value of x ?

Answer

- A. $-\frac{76}{5}$
- B. -9
- C. 5
- D. 76

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$7m = 5(n + p)$

The given equation relates the positive numbers m , n , and p . Which equation correctly gives n in terms of m and p ?

Answer

- A. $n = \frac{5p}{7m}$
- B. $n = \frac{7m}{5} - p$
- C. $n = 5(7m) + p$
- D. $n = 7m - 5 - p$

Question ID: 137cc6fd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$$\sqrt[5]{70n}\left(\sqrt[3]{70n}\right)^2$$

For what value of x is the given expression equivalent to $(70n)^{30x}$, where $n > 1$?

Question ID: 6ce95fc8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$2x^2 - 2 = 2x + 3$

Which of the following is a solution to the equation above?

Answer

- A. 2
- B. $1 - \sqrt{11}$
- C. $\frac{1}{2} + \sqrt{11}$
- D. $\frac{1 + \sqrt{11}}{2}$

Question ID: 4dd4efcf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$f(x) = ax^2 + 4x + c$

In the given quadratic function, a and c are constants. The graph of $y = f(x)$ in the xy -plane is a parabola that opens upward and has a vertex at the point (h, k) , where h and k are constants. If $k < 0$ and $f(-9) = f(3)$, which of the following must be true?

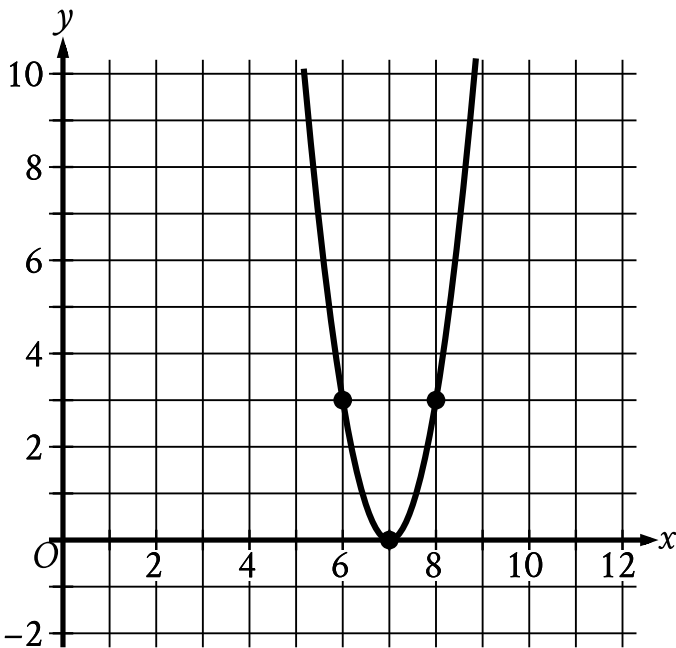
- I. $c < 0$
- II. $a \geq 1$

Answer

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question



The x -intercept of the graph shown is $(x, 0)$. What is the value of x ?

Question ID: 6bdcac03

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$x^2 = -841$

How many distinct real solutions does the given equation have?

Answer

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

Question ID: f5aa5040

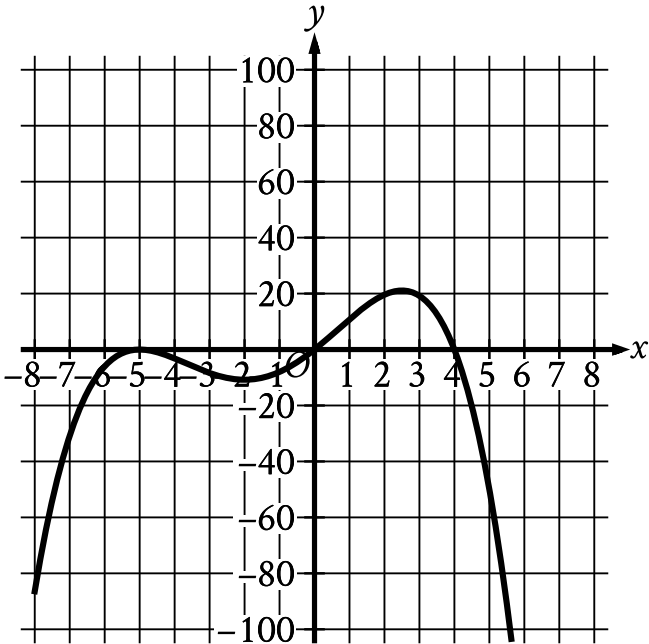
Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

In the xy -plane, a line with equation $2y = c$ for some constant c intersects a parabola at exactly one point. If the parabola has equation $y = -2x^2 + 9x$, what is the value of c ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question



Which of the following could be the equation of the graph shown in the xy -plane?

Answer

- A. $y = -\frac{1}{10}x(x - 4)(x + 5)$
- B. $y = -\frac{1}{10}x(x - 4)(x + 5)^2$
- C. $y = -\frac{1}{10}x(x - 5)(x + 4)$
- D. $y = -\frac{1}{10}x(x - 5)^2(x + 4)$

Choice C is incorrect. The graph of this equation has x-intercepts at $x = -4$, $x = 0$, and $x = 5$, rather than x-intercepts at $x = -5$, $x = 0$, and $x = 4$.

Choice D is incorrect. The graph of this equation has x-intercepts at $x = -4$, $x = 0$, and $x = 5$, rather than x-intercepts at $x = -5$, $x = 0$, and $x = 4$.